

INTERRUPT ROUTINE FOR 64 x 1 FORMAT (8 BYTE MEM)

<u>MACHINE CODE</u>	<u>ASSEMBLY LANGUAGE</u>	<u>COMMENTS</u>
72	INTRET: LDXA	..RESTORE D
70	RET	..RETURN
C4	INT : NOP	..SYNCHRONIZING INSTR
22	DEC R2	..R2=STACK PTR
78	SAV	..T→ STACK
22	DEC R2	
52	STR R2	..D→ STACK
E2 E2 E2 E2	SEXR2; SEXR2; SEXR2; SEXR2	..NOPS
F8 ____ B0	A.1(DISMEM)→R0.1	..LOAD R0 WITH
F8 ____	A.0(DISMEM)→ &	..START ADDR OF DISP MEM
A0]	DISP : PLO R0	..LINE ADDR R0.0
20	DEC R0	..KEEPS DMA PTR ON DISPLAY PG
3C ____	BN1 DISP	
A0	DISPEF: PLO R0	..LAST 4 LINES (EF1 = 1)
20	DEC R0	
34 ____	B1 DISPEF	
30	BR INTRET	..END OF DISPLAY

Figure 6

INTERRUPT ROUTINE FOR 64 x 32 FORMAT (1 PG MEM)

<u>MACHINE CODE</u>	<u>ASSEMBLY LANGUAGE</u>	<u>COMMENTS</u>
72	INTRET: LDXA	..RESTORE D
70	RET	..RETURN
C4	INT : NOP	..3 CYC. INSTR. USED ..FOR PGM. SYNC
22	DEC R2	..R2 IS STACK PTR
78	SAV	..T → STACK
22	DEC R2	
52	STR R2	..D → STACK
F8 ____ B0	A.1(DISMEN) → RO.1	..LOAD RO WITH
F8 ____ A0	A.0(DISMEN) → RO.0	..START. ADDR. OF DISP. MEM
E2 E2 E2 E2	SEX2; SEX2; SEX2;SEX2	..NOPS USED FOR SYNC
80]	DISP : GLO RO	..LINE START ADDR. → D
E2	SEX2	..NOP
20	DEC RO	..RESET RO.1 IF PASS PG
A0]	PLO RO	..LINE START ADDR. → RO.0
E2	SEX2	..NOP
20	DEC 20	..RESET RO.1 IF PASS PG
A0]	PLO RO	..LINE START ADDR. → RO.0
E2	SEX2	..NOP
20	DEC RO	..RESET RO.1 IF PASS PG
A0]	PLO RO	..REPEATS SAME LINE
E2	SEX2	..NOP
3C ____	BN1 DISP	..LOOPS 32 TIMES
30 ____	BR INTRET	..END OF DISPLAY

Figure 7

INTERRUPT ROUTINE FOR 64 x 64 FORMAT (2 PGS MEM)

<u>MACHINE CODE</u>	<u>ASSEMBLY LANGUAGE</u>	<u>COMMENTS</u>
72	INTRET: LDXA	..RESTORE D
70	RET	..RETURN
C4	INT : NOP	..3 CYC. INSTR. FOR PGM. SYNC
22	DEC R2	..R2 IS STACK PTR
78	SAV	..T → STACK
22	DEC R2	
52	STR R2	..D → STACK
F8 ____ B0	A.1(DISEM) → RO.1	..DISEM IS START ADDR
F8 ____ A0	A.0(DISEM) → RO.0	..OF DISPLAY MEMORY
E2 E2 E2 E2	SEX2; SEX2; SEX2; SEX2	..NOPS FOR PGM SYNC
80]	DISP : GLO RO	..NEW LINE
E2	SEX2	..NOP
20	DEC RO	..RESTORES RO.1 IF PASS PG
A0]	PLO RO	..REPEATS SAME LINE
E2	SEX2	..NOP
3C ____	BN1 DISP	..LOOP 60 TIMES
80]	DISEF: GLO RO	..LAST 4 VIDEO LINES
E2	SEX RO	..NOP
20 A0]	DEC RO; PLO RO	
E2	SEX2	..NOP
34 ____	B1 DISEF	
30 ____	BR INTRET	..END OF DISPLAY

Figure 8

INTERRUPT ROUTINE FOR 64 x 128 (4 PGS MEM)

<u>MACHINE CODE</u>	<u>ASSEMBLY LANGUAGE</u>	<u>COMMENTS</u>
72	INTRET: LDXA	..RESTORE D
70]	RET	..RETURN
C4	INT : NOP	..ENTRY POINT
22	DEC R2	..R2 = STACK PTR
78	SAV	..T → STACK
22	DEC R2	
52	STR R2	..D → STACK
E2 E2	SEX R2; SEX R2	..NOP
F8 ____ B0	A.1(DISEM) → RO.1	..LOAD RO WITH
F8 ____ A0	A.0(DISEM) → RO.0	..START ADDR OF DISP. MEM.
30 ____	BR INTRET	

Figure 9

GAMES FOR FRED III

SPLIT-SECOND

EACH TIME PATTERN IS SHOWN, FIRST PLAYER TO
MATCH ALL THE FLASHING SPOTS AT SAME TIME
SCORES, FIRST PLAYER TO MAKE A SCORE OF
100 WINS.

CHASE

FOR TWO PLAYER GAME, PRESS * FOR PLAYER B,
AND FOR SINGLE PLAYER GAME, PRESS * FOR
PLAYER A,

SPACE SHIP FOR PLAYER A STARTS AT LEFT, AND
THAT FOR PLAYER B (WITH HOLE IN SYMBOL)
STARTS AT RIGHT, BOTH VESSELS CAN BE MOVED
IN DIRECTIONS OF ARROWS FOR EACH PLAYER,
A PLAYER SCORES WHEN HIS VESSEL CAPTURES
THE OPPONENTS VESSEL, THE FIRST PLAYER
TO REACH A SCORE OF 50 WINS GAME,

MINES

FOR TWO PLAYER GAME, PRESS * FOR PLAYER B,
AND FOR SINGLE PLAYER GAME, PRESS * FOR
PLAYER A,

SPACE SHIP FOR PLAYER A STARTS AT LEFT AND
THAT FOR PLAYER B (WITH HOLE IN SYMEOL)
STARTS AT RIGHT. PLAYER SCORES WHEN OPPONENT'S
VESSEL IS ENTRAPPED BY MINES OR OTHER VESSEL,
FIRST PLAYER TO REACH A SCORE OF 50 WINS GAME,
MINES CAN BE LAID BY BOTH VESSELS WHEN EITHER
PLAYER PRESSES HIS * BUTTON,

NOTE: SPEED OF GAME IS CONTROLLED BY POSITION
OF FIRST TWO SWITCHES UNDER BODY OF GAME:

SWITCHES	1	2	3	4
FAST	OFF	OFF	-	-
FAST	ON	OFF	-	-
FAST	OFF	ON	-	-
SLOW	ON	ON	-	-

BOWLING

FOR EACH PLAYER THE BALL IS RELEASED BY THE FOLLOWING
BUTTONS:

↑ - HOOK UP

* - STRAIGHT BALL

↓ - HOOK DOWN

OTHER BUTTONS ARE NOT USED IN THIS GAME.

RACING

TO SEE TOTE BOARD PRESS EITHER * BUTTONS,
CHOOSE HORSE (1-4) WITH NUMBERED BUTTON,
THEN PLACE BET BY HOLDING DOWN RESPECTIVE
* UNTIL AMOUNT DESIRED IS REACHED, ODDS
GIVEN ON TOTE BOARD ARE REALISTIC ODDS
THAT ANY GIVEN HORSE WILL WIN, LOSERS
WILL LOSE BET FROM CASH ON HAND AND WINNERS
WILL RECEIVE (AMOUNT BET) X (ODDS),

MAZE-DRAW

MAZE

TO PLAY MAZE, HOLD BOTH * BUTTONS DOWN, PRESS AND RELEASE START, AND THEN RELEASE * BUTTONS, OBJECT IS FOR BOTH PLAYERS TO REACH STARTING POSITION OF OTHER PLAYER FIRST, END TARGET SHOWN AND OBTAINED BY PRESSING * BUTTON.

DRAW

THE STARTING MAZE USED ABOVE CAN BE CHANGED OR REPLACED BY DRAWING YOUR OWN.

TO CHANGE EXISTING MAZE, HOLD JUST * FOR PLAYER B AND PRESS AND RELEASE START, NOW YOU ARE NO LONGER IN MAZE MODE, PATTERNS CAN BE ADDED OR REMOVED BY PRESSING * BUTTON WHILE MOVING WITH ARROWS.

TO ERASE ALL EXISTING PATTERNS AND CREATE NEW ONES, HOLD JUST * FOR PLAYER A AND PRESS AND RELEASE START.

MAZE

TO PLAY MAZE WITH YOUR NEW PATTERN, JUST PRESS
AND RELEASE START (WITHOUT PRESSING ANY OTHER
BUTTONS),

NOTE: SPEED OF MAZE IS DETERMINED BY POSITION
OF FIRST TWO SWITCHES UNDER BODY OF GAME:

SWITCHES	1	2	3	4
FAST	OFF	OFF	-	-
FAST	ON	OFF	-	-
FAST	OFF	ON	-	-
SLOW	ON	ON	-	-

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MAZE - DRAW

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